27. GPTW Port Output Enable (POEG)

27.1 Overview

The POEG issues requests to stop output from output pins of the general PWM timer (GPTW). The combination of output pins of the POEG to be stopped can be specified from any channel of the GPTW.
Select the method of detection for stopping the output from the list below.

- Input level detection on the GTETRGn pin (n = A to D)
- Detection from the GPTW to stop output
- Detection of stopping of oscillation by the oscillation stop detection circuit for the main clock
- Register setting

The GTETRGn pin can be used for output to the external trigger input pins of the GPTW.
Table 27.1 shows specifications, Figure 27.1 shows a schematic view of the system, Figure 27.2 shows a block diagram, and Table 27.2 shows input pins.

Table 27.1 POEG Specifications

Item	Description
Request of output stopping in response to input level detection	• The request to stop output is issued to the GPTW when a POEGGn.PIDF flag is set in response to the detection of input of the selected level on the corresponding GTETRGn pin (n = A to D).
Requests to stop output in the form of an output stopping signal from the GPTW	• The request to stop output is issued to the GPTW when the GPTW detects the active level (high or low) on the GTIOCA and GTIOCB pins at the same time while the corresponding POEGGn.IOCF flag is set. • The request to stop output is issued to the GPTW when the GPTW detects a dead-time error while the corresponding POEGGn.IOCF flag is set.
Requests to stop output in response to detecting the stopping of oscillation	A request to stop output is issued to the GPTW when the oscillation stop detection circuit for the main clock detects the stopping of oscillation while the corresponding POEGGn.OSTPF flag is set.
Requests by software to stop output	The request to stop output is issued to the GPTW when the software sets the POEGGn.SSF flag.
Interrupt	• An interrupt is generated in response to the request to stop output in the form of the POEGGn.PIDF flag. • An interrupt is generated in response to the request to stop output in the form of the POEGGn.IOCF flag.
External trigger output to the GPTW	The GTETRGn pin is used for output as external triggers.
Noise removal	• Each GTETRGn pin has a digital noise filter. • Four types of sampling clock are selectable for the filter.

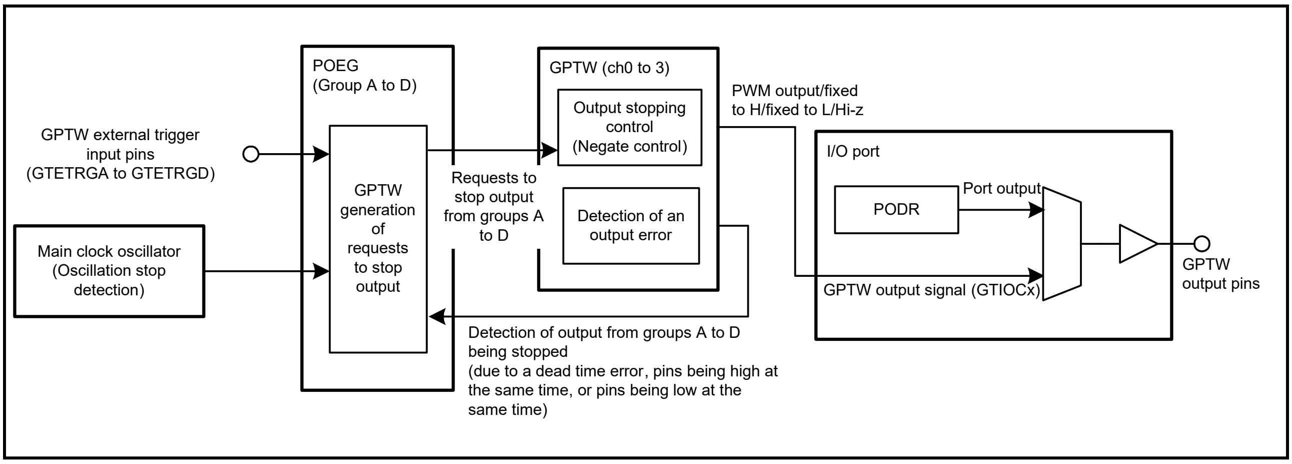

Figure 27.1 Schematic View of the POEG System

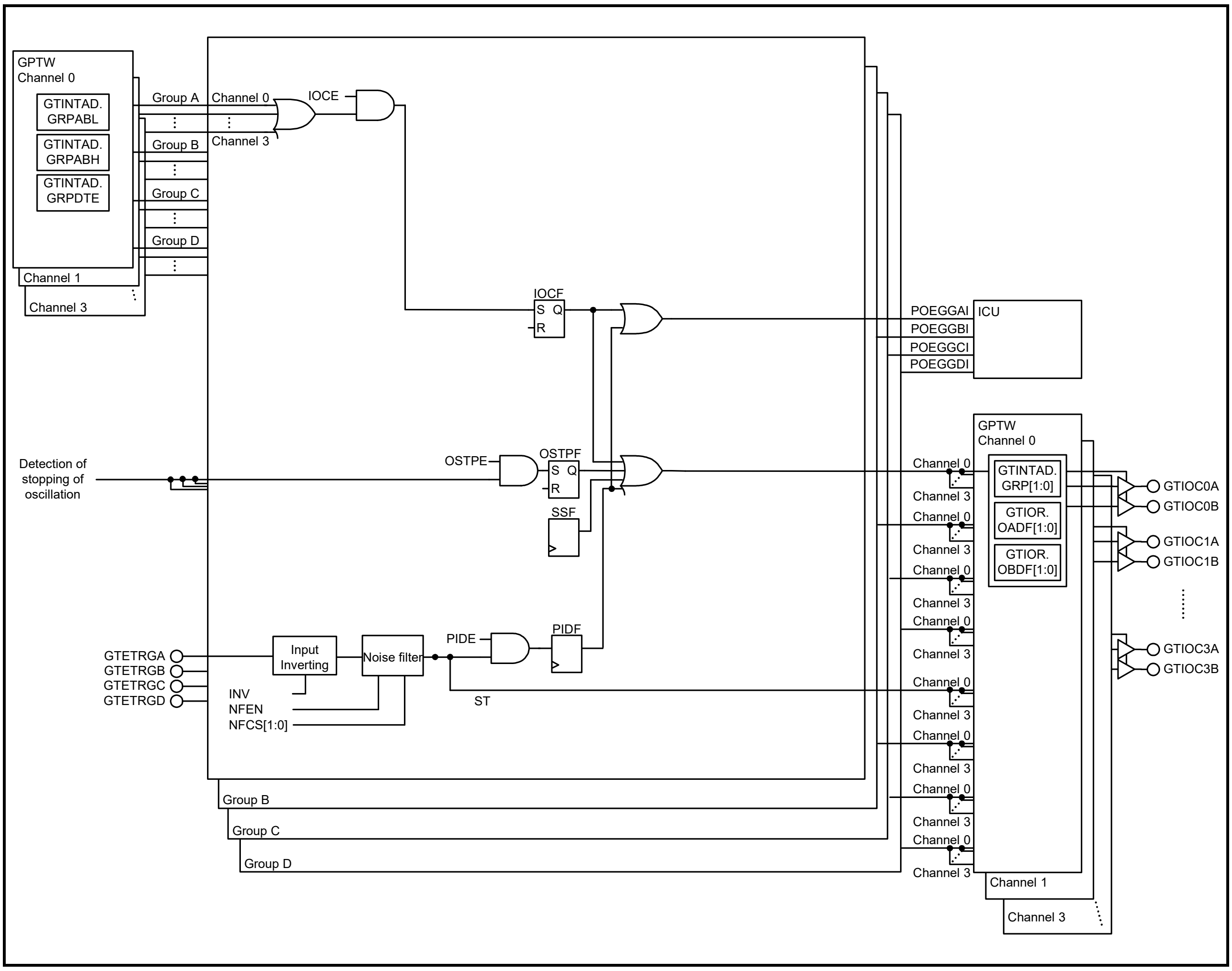

Figure 27.2 POEG Block Diagram

Table 27.2 POEG I/O Pins

Pin Name	I/O	Function
GTETRGA	Input	Output stop request signal for GPTW output pin and GPTW external trigger input pin A
GTETRGB	Input	Output stop request signal for GPTW output pin and GPTW external trigger input pin B
GTETRGC	Input	Output stop request signal for GPTW output pin and GPTW external trigger input pin C
GTETRGD	Input	Output stop request signal for GPTW output pin and GPTW external trigger input pin D

27.2 Register Descriptions

27.2.1 POEG Group n Setting Register (POEGGn) (n = A to D)

Address(es): POEG.POEGGA 0009 E000h, POEG.POEGGB 0009 E100h, POEG.POEGGC 0009 E200h, POEG.POEGGD 0009 E300h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	NFCS[1:0]		NFEN	INV	—	—	—	—	—	—	—	—	—	—	—	ST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	OSTPE	IOCE	PIDE	SSF	OSTPF	IOCF	PIDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PIDF	Port Input Detection Flag	0: The selected input level was not detected on the GTETRGn pin. 1: The selected input level was detected on the GTETRGn pin.	R/(W)*1
b1	IOCF	GPTW Output Stop Request Detection Flag	0: Stopping of GPTW output was not detected. 1: Stopping of GPTW output was detected.	R/(W)*1
b2	OSTPF	Oscillation Stop Detection Flag	0: Stopping of oscillation was not detected. 1: Stopping of oscillation was detected.	R/(W)*1
b3	SSF	Software Stop Flag	0: Software has not stopped output. 1: Software has stopped output.	R/W
b4	PIDE	Port Input Detection Enable	0: Detection of input levels on the corresponding GTETRGn pin is disabled. 1: Detection of input levels on the corresponding GTETRGn pin is enabled.	R/W*2
b5	IOCE	GPTW Output Stop Request Enable	0: Detection of stopping of output from the GPTW is disabled. 1: Detection of stopping of output from the GPTW is enabled.	R/W*2
b6	OSTPE	Enable Stopping Output on Stopping of Oscillation	0: Detection of stopping of oscillation is disabled. 1: Detection of stopping of oscillation is enabled.	R/W*2
b15 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b16	ST	GTETRGn Input Status Flag	0: The corresponding external trigger for output to the GPTW is 0. 1: The corresponding external trigger for output to the GPTW is 1.	R
b27 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b28	INV	GTETRGn Input Inverting	0: Input on the GTETRGn pin is not inverted. 1: Input on the GTETRGn pin is inverted.	R/W
b29	NFEN	Noise Filter Enable	0: Digital noise filter on the GTETRGn pin is disabled. 1: Digital noise filter on the GTETRGn pin is enabled.	R/W
b31, b30	NFCS[1:0]	Noise Filter Clock Select	b31 b30 0 0: Samples the input level of GTETRGn pin three times per PCLKB/1 clock 0 1: Samples the input level of GTETRGn pin three times per PCLKB/8 clock 1 0: Samples the input level of GTETRGn pin three times per PCLKB/32 clock 1 1: Samples the input level of GTETRGn pin three times per PCLKB/128 clock	R/W

Note 1. Only 0 to clear the flag can be written.
Note 2. This bit can be written only once after resetting.

The POEGGn register (n = A to D) controls requests for stopping output and an external trigger for the GPTW based in response to the detection of various signals.

SSF Flag (Software Stop Flag)

Writing 1 to the SSF flag leads to a request to stop output being issued to the GPTW, and writing 0 to the flag releases the GPTW from the request to stop output. In addition, requests to stop output issued by software can be monitored by reading the flag.

27.3 Operation

27.3.1 Request to Stop Output in Response to Detection of Input Level on the Corresponding GTETRGn Pin (n = A to D)

Output stopping request: requests in response to setting of the POEGGn.PIDF flag (n = A to D).
The input set in the POEGGn register (inversion or non-inversion set in the INV bit, filtering enabled or disabled by the NFEN bit, sample clock for filtering set in the NFCS[1:0] bits) is detected while the POEGGn.PIDE bit is 1. After the POEGGn.PIDF flag is set to 1 in response to this, the request to stop output is issued per group to each channel of the GPTW. To de-assert the request signal for stopping output, clear the POEGGn.PIDF flag. For canceling requests to stop output, refer to section 27.3.5, Canceling Requests to Stop Output.

27.3.1.1 Digital Noise Filter

Each GTETRGn pin input has a digital noise filter. Figure 27.3 shows the operation example in detection of the high level with the use of the filter. When the digital noise filter is enabled (POEGGn.NFEN bit = 1) and the high level is detected 3 consecutive times with the sampling clock cycle selected by the POEGGn.NFCS[1:0] bits while inverted or non-inverted according to the setting of the POEGGn.INV bit, this is regarded as detection of the high level, and the request to stop output is issued to the GPTW.
At this time, if the low level is detected even once in the sequence, it is not regarded as detection of the high level. In addition, changes in the levels on pins GTETRGA through GTETRGD are ignored while the sampling clock is not being output.
Digital noise filters can be used with requests to stop output in response to setting of the POEGGn.PIDF flags (n = A to D) and the output of external triggers to the GPTW.

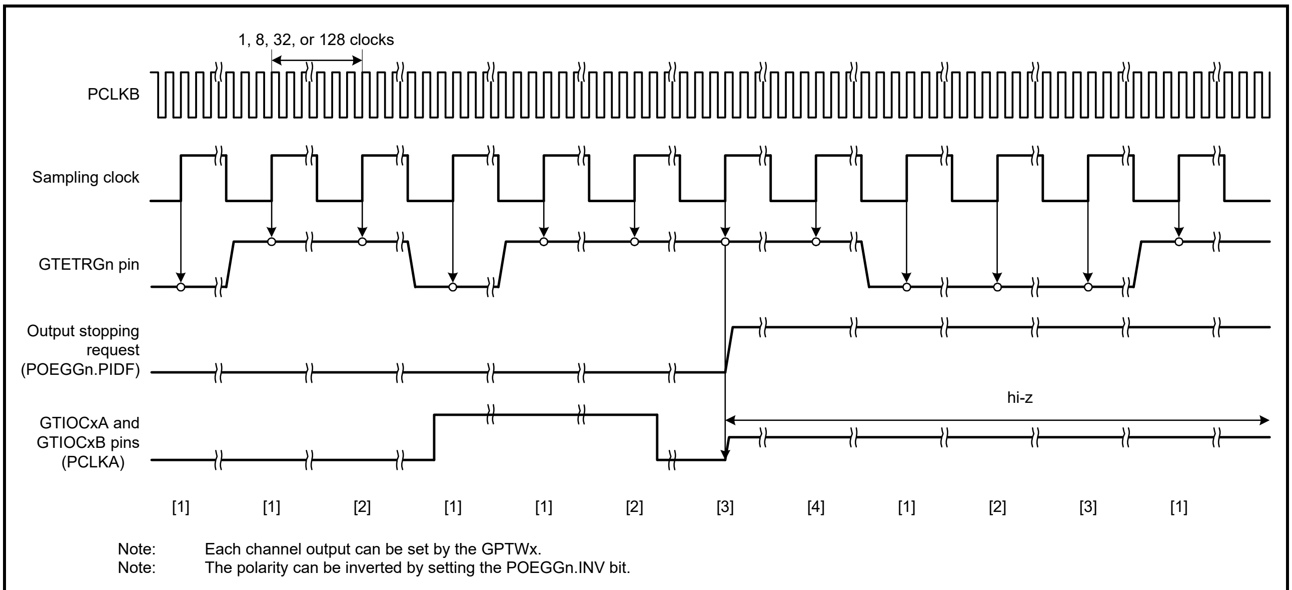

Figure 27.3 Example for Operation of Digital Noise Filter

27.3.2 Requests to Stop Output in Response to Detection of Output Stopping form GPTW

When any among a dead-time error or simultaneous high- or low-level output from the GPTW is detected, the corresponding POEGGn.IOCF flag is set to 1 and the request to stop output per group is issued to each channel in the GPTW. To cancel a request to stop output, clear the given POEGGn.IOCF flag. For details, refer to section 27.3.5, Canceling Requests to Stop Output.

To detect any among a dead-time error or simultaneous high- or low-level output from the GPTW, detection of output stopping must be permitted in the GRPDTE, GRPABH, and GRPABL bits of the given GPTWn.GTINTAD register. Specify the group of the GPTW for which stopping is to be detected in the GPTWn.GTINTAD.GRP[1:0] bits. For details, refer to section 26.2.15, General PWM Timer Interrupt Output Setting Register (GTINTAD).

27.3.3 Requests to Stop Output by Oscillation Stop Detection

When the circuit for detecting stopping of oscillation by the main clock oscillator detects the stopping of the oscillation while a POEGGn.OSTPE bit is 1, the POEGGn.OSTPF flag is set to 1 and a request to stop output per group is issued to each channel of the GPTW. To cancel the request to stop output, clear the POEGGn.OSTPF flag. For details, refer to section 27.3.5, Canceling Requests to Stop Output.

27.3.4 Requests to Stop Output by a Register

Writing 1 to the software stop flag (POEGGn.SSF) leads to a request to stop output per group to each channel in the GPTW. To release from the request to stop output, clear the POEGGn.SSF flag. For details, refer to section 27.3.5, Canceling Requests to Stop Output.

27.3.5 Canceling Requests to Stop Output

Requests to stop output are canceled in any of the following two ways.

(1) Cancellation by a reset (return to the initial state)

(2) Cancellation by clearing all flags in the POEGGn register

(1) Cancellation by a reset

Any type of reset can cancel a request to stop output. For details, refer to section 6, Resets.

(2) Cancellation by clearing all flags in the POEGGn register

- POEGGn.PIDF
- POEGGn.IOCF
- POEGGn.OSTPF
- POEGGn.SSF

Cancellation of the request is taken into the GPTW at the end of the cycle of counting by the GPTW, and the output pins are released from being stopped no less than 3 PCLKA cycles from that time. Figure 27.4 shows the timing of release from the output-stopped state. To clear each of the flags, read the status flag for each source to check that the source condition is not being detected, and then write 0. Sources cannot be cleared even if the flag is cleared in the detected state. Status flags for the respective sources are listed below.

- Detection of input level — POEGGn.ST (GTETRGn input status flag)
- Oscillation stop detection — OSTDSR.OSTDF (oscillation stop detection flag)
- Stopping detection from the GPTW — GPTWn.GTST.DTEF (dead time error flag)
  GPTWn.GTST.OABLF (simultaneous low output flag)
  GPTWn.GTST.OABHF (simultaneous high output flag)

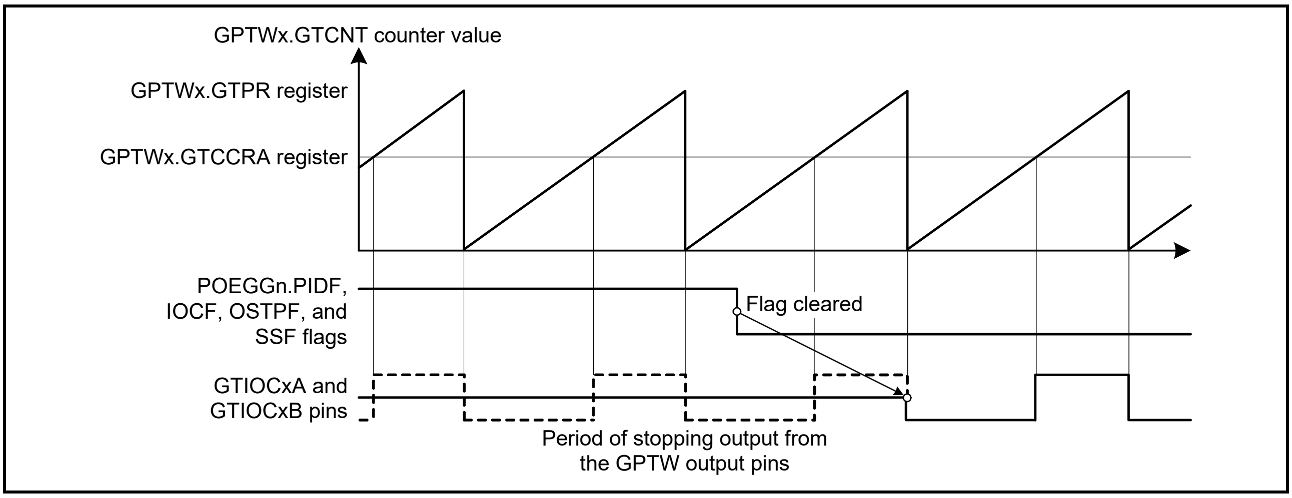

Figure 27.4 Timing of Re-enabling Output from the GPTW Output Pins Following Cancellation of a Request to Stop Output

27.4 Interrupt Sources

The POEG generates interrupts for the interrupt controller when any of the following is detected.

- Detection of level of input, indicated by the POEGGn.PIDF flag
- Detection of stopping of output from the GPTW, indicated by the POEGGn.IOCF flag

Table 27.3 shows interrupt sources and conditions.

Table 27.3 Interrupt sources and conditions

Interrupt Source	Symbol	Corresponding Flag	Trigger Condition
POEG Group A Interrupt	POEGGAI	POEGGA.IOCF	Detection of stopping of output from the GPTW
		POEGGA.PIDF	Detection of the level input from the GTETRGA pin
POEG Group B Interrupt	POEGGBI	POEGGB.IOCF	Detection of stopping of output from the GPTW
		POEGGB.PIDF	Detection of the level input from the GTETRGB pin
POEG Group C Interrupt	POEGGCI	POEGGC.IOCF	Detection of stopping of output from the GPTW
		POEGGC.PIDF	Detection of the level input from the GTETRGC pin
POEG Group D Interrupt	POEGGDI	POEGGD.IOCF	Detection of stopping of output from the GPTW
		POEGGD.PIDF	Detection of the level input from the GTETRGD pin

27.5 External Trigger Output to GPTW

The POEG outputs the inputs on the GTETRGn pin (n = A to D) as external trigger signals. The external trigger signals can be monitored by the POEGGn.ST flag via active-sense selection and digital noise filtering. The GPTW can function as follows in response to external trigger signals.

- Count start
- Count stop
- Counter clearing
- Up-counting
- Down-counting
- Input capture

For details of the above functions, refer to section 26, General PWM Timer (GPTW).

27.6 Usage Notes

27.6.1 Transitions to Low-Power Consumption Mode

When the POEG is used, it should not be transitioned to software standby or deep software standby modes. In this mode, the stopping of output cannot be requested since the POEG is stopped.

27.6.2 Setting the Function for Stopping the Module

The module-stop control register can be set to enable or disable operation of the POEG. The POEG is stopped immediately after a reset. Registers of the POEG become accessible following release from the module-stopped state. For details, refer to section 11, Low Power Consumption. The module-stop bit for the POEG also stops and starts the GPTW.

27.6.3 Duplication of Requests to Stop Output

Note that requests to stop output will not be canceled when a corresponding flag is set stop output in response to detection. Request signals to stop output in response to flag setting are obtained as the logical OR of the corresponding detection signals for stopping output.